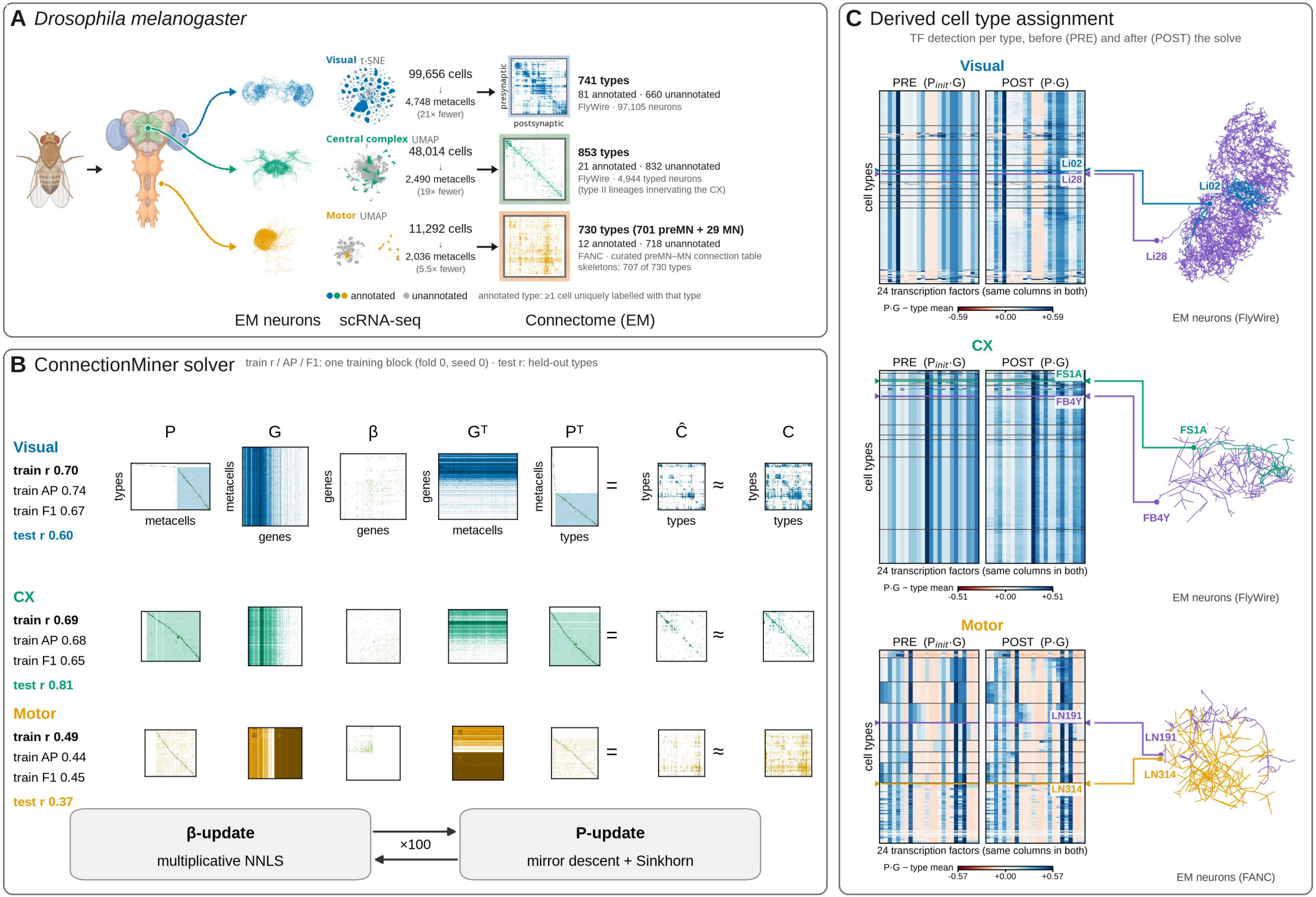

A Drosophila melanogaster
Visual t-SNE
99,656 cells
4,748 metacells
(21× fewer)
presynaptic
postsynaptic
741 types
81 annotated · 660 unannotated
FlyWire · 97,105 neurons
Central complex UMAP
48,014 cells
2,490 metacells
(19× fewer)
853 types
21 annotated · 832 unannotated
FlyWire · 4,944 typed neurons
(type II lineages innervating the CX)
Motor UMAP
11,292 cells
2,036 metacells
(5.5× fewer)
730 types (701 preMN + 29 MN)
12 annotated · 718 unannotated
FANC · curated preMN–MN connection table
skeletons: 707 of 730 types
annotated
unannotated
annotated type: ≥1 cell uniquely labelled with that type
EM neurons
scRNA-seq
Connectome (EM)
B ConnectionMiner solver
train r / AP / F1: one training block (fold 0, seed 0) · test r: held-out types
P
G
β
G^T
P^T
Ĉ
C
Visual
train r 0.70
train AP 0.74
train F1 0.67
test r 0.60
types
metacells
genes
CX
train r 0.69
train AP 0.68
train F1 0.65
test r 0.81
Motor
train r 0.49
train AP 0.44
train F1 0.45
test r 0.37
β-update
multiplicative NNLS
×100
P-update
mirror descent + Sinkhorn
C Derived cell type assignment
TF detection per type, before (PRE) and after (POST) the solve
Visual
PRE (P_init·G)
POST (P·G)
cell types
Li02
Li28
24 transcription factors (same columns in both)
P·G − type mean
-0.59
+0.00
+0.59
EM neurons (FlyWire)
CX
FS1A
FB4Y
-0.51
+0.00
+0.51
EM neurons (FlyWire)
Motor
LN191
LN314
-0.57
+0.00
+0.57
EM neurons (FANC)